Probability – Problems

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Spring 2026

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2 Laws of large numbers

2.2 Weak laws of large numbers

Exercise (Durrett 2.2.1)**.** Let $X_1, X_2, \dots$ be uncorrelated random variables with finite expectations and

$$i^{-1} \operatorname{var} X_i \to 0 \text{ as } i \to \infty.$$

Let $S_n = X_1 + \cdots + X_n$ and $\mu_n = ES_n/n$. Show that $S_n/n - \mu_n \to 0$ in probability as $n \to \infty$.

Proof. WLOG assume $EX_n = 0$ for all n. Now we just need to show that $S_n/n \to 0$ in probability. Fix $\delta > 0$, we next show

$$P\left(\left|\frac{S_n}{n}\right| > 2\delta\right) \to 0$$

Let $\varepsilon > 0$ be arbitrary. Since $i^{-1} \operatorname{var} X_i \to 0$ as $i \to \infty$, there exists N such that $i \geq N$ implies $i^{-1} \operatorname{var} X_i < \varepsilon$. Let $n \geq N$, we have

$$\begin{aligned} P\left(\left|\frac{S_n}{n}\right| > 2\delta\right) &= P\left(\frac{1}{n}\left|\sum_{i=1}^{n} X_i\right| > 2\delta\right) \\ &\leq P\left(\frac{1}{n}\sum_{i=1}^{N} |X_i| + \frac{1}{n}\left|\sum_{i=N+1}^{n} X_i\right| > 2\delta\right) \\ &\leq P\left(\frac{1}{n}\sum_{i=1}^{N} |X_i| > \delta\right) + P\left(\frac{1}{n}\left|\sum_{i=N+1}^{n} X_i\right| > \delta\right) \end{aligned}$$

For the first term, Chebyshev's inequality gives

$$P\left(\frac{1}{n}\sum_{i=1}^{N} |X_i| > \delta\right) \leq \frac{1}{n\delta}\sum_{i=1}^{N} E\left|X_i\right| \to 0$$

as $n \to \infty$. For the second term, Chebyshev's inequality gives

$$P\left(\frac{1}{n}\left|\sum_{i=N+1}^{n} X_i\right| > \delta\right) \leq \frac{1}{\delta^2} \operatorname{var}\left(\frac{1}{n}\sum_{i=N+1}^{n} X_i\right) = \frac{1}{n^2\delta^2} \operatorname{var}\left(\sum_{i=N+1}^{n} X_i\right).$$

Since X_i's are uncorrelated, it follows that

$$P\left(\frac{1}{n}\left|\sum_{i=N+1}^{n} X_i\right| > \delta\right) \leq \frac{1}{n^2\delta^2}\sum_{i=N+1}^{n} \operatorname{var}(X_i) \leq \frac{1}{n\delta^2}\sum_{i=N+1}^{n} \frac{\operatorname{var} X_i}{i} \leq \frac{(n-N)\varepsilon}{n\delta^2} \leq \frac{\varepsilon}{\delta^2}.$$

Since $\varepsilon > 0$ is arbitrary, this completes the proof.

$\square$

Exercise (Durrett 2.2.2)**.** The L^2 weak law generalizes immediately to certain dependent sequences. Suppose EX_n and $EX_nX_m \le r(n-m)$ for $m \le n$ (no absolute value on the left-hand side!) with $r(k) \to 0$ as $k \to \infty$. Show that $(X_1 + \cdots + X_n)/n \to 0$ in probability.

Proof. Consider

$$\begin{aligned} E\left(n^{-1}\left(X_1 + \cdots + X_n\right)\right)^2 &= n^{-2} E\left(X_1 + \cdots + X_n\right)^2 \\ &= n^{-2}\left(\sum_{i=1}^{n} EX_i^2 + 2\sum_{i<j} EX_i E_j\right) \\ &\le n^{-2}\left(nr(0) + 2\sum_{i=1}^{n-1}(n-i)r(i)\right). \end{aligned}$$

Let ε be arbitrary. Since $r(k) \to 0$ as $k \to \infty$, there is N such that $r(k) \le \varepsilon$ for all $k \ge N$. Hence for $n \ge N$, we have

$$\begin{aligned} E\left(n^{-1}(X_1 + \cdots + X_n)\right)^2 &\le \frac{r(0)}{n} + \frac{2}{n^2}\sum_{i=1}^{N}(n-i)r(i) + \frac{2}{n^2}\sum_{i=N+1}^{n}(n-i)\varepsilon \\ &\le \frac{r(0)}{n} + \frac{2N}{n}\sum_{i=1}^{N} r(i) + 2\varepsilon. \end{aligned}$$

This shows that $(X_1 + \cdots + X_n)/n \to 0$ in L^2 and thus in probability.

□

Exercise (Durrett 2.2.5)**.** Let $X_1, X_2, \ldots$ be i.i.d. with

$$P(X_i > x) = \frac{e}{x \log x}.$$

for $x \ge e$. Show that $E|X_i| = \infty$, but there is a sequence of constants $\mu_n \to \infty$ so that $S_n/n - \mu_n \to 0$ in probability.

Proof. Note that $P(X_i < e) = 0$. It follows that

$$E|X_i| = \int_e^\infty P(X_i > x)\,dx = [\log\log x]_e^\infty = \infty.$$

However,

$$xP(X_i > x) = \frac{e}{\log x} \to 0.$$

Hence, by a previous theorem, if $\mu_n = EX_1 \mathbf{1}_{(|X_1| \le n)}$ then $S_n/n - \mu_n \to 0$. Note that $\mu_n < \infty$ but $\mu_n \to EX_i = \infty$ as $n \to \infty$ by monotone convergence theorem.

□

2.3 Borel-Cantelli lemmas

Exercise (Durrett 2.3.8)**.** Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of independent events with $P(A_n) < 1$ for all n. Show that $P(\bigcup_{n=1}^{\infty} A_n) = 1$ implies $\sum_{n=1}^{\infty} P(A_n) = \infty$ and hence $P(A_n \text{ i.o.}) = 1$.

Proof. Suppose for contradiction that $\sum_{n=1}^{\infty} P(A_n) = M < \infty$. From independence, we have

$$P\left(\bigcap_{n=1}^{N} A_n^c\right) = \prod_{n=1}^{N}(1 - P(A_n)).$$

As $\sum_{n=1}^{\infty} P(A_n)$ converges, $P(A_n) \to 0$ as $n \to 0$. It follows that there exists $\alpha < 1$ such that $P(A_n) \leq \alpha < 1$ for all n. Once we select $\beta > 0$ such that $1 - \alpha \geq \exp(\beta\alpha)$, we obtain

$$P\left(\bigcap_{n=1}^{N} A_n^c\right) = \prod_{n=1}^{N}(1 - P(A_n)) \geq \exp\left(-\beta \sum_{n=1}^{N} P(A_n)\right)$$

This implies

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = 1 - P\left(\bigcap_{n=1}^{\infty} A_n^c\right) \leq 1 - \exp\left(-\beta \sum_{n=1}^{\infty} P(A_n)\right)$$

Hence, $P(\bigcup_{n=1}^{\infty} A_n) \leq 1 - \exp(-\beta M) < 1$, contradiction.

□

Exercise (Durrett 2.3.9)**.** Show that if $P(A_n) \to 0$ and $\sum_{n=1}^{\infty} P(A_n^c \cap A_{n+1}) < \infty$, then $P(A_n \text{ i.o.}) = 0$. Find an example of a sequence $\{A_n\}_{n=1}^{\infty}$ to which the result above can be applied but the Borel-Cantelli lemma cannot.

Proof. For any N, M, we have

$$\bigcup_{n=N}^{M} A_n = A_N \cup \bigcup_{n=N}^{M-1} (A_n^c \cap A_{n+1})$$

This implies

$$P\left(\bigcup_{n=N}^{M} A_n\right) \leq P(A_N) + \sum_{n=N}^{M-1} P(A_n^c \cap A_{n+1}).$$

Letting $M \to \infty$ shows $P(\bigcup_{n=N}^{\infty} A_n) \leq P(A_n) + \sum_{n=N}^{\infty} P(A_n^c \cap A_{n+1})$. Now letting $N \to \infty$ shows $P(\limsup A_n) = 0$, as desired.

For an example, define $(\Omega, \mathcal{F}, P)$ by letting $\Omega = [0, 1]$, $\mathcal{F}$ be the Borel sets, and P be the Lebesgue measure. For $n \geq 1$, define A_n via

$$A_n = \left[1 - 2^{-n} - n^{-1}, 1 - 2^{-n}\right].$$

Then $P(A_n) = n^{-1}$ but $P(A_n^c \cap A_{n+1}) = 2^{-n-1}$.

□

Exercise (Durrett 2.3.12)**.** Let $X_1, X_2, \dots$ be a sequence of r.v.'s on $(\Omega, \mathcal{F}, P)$ where Ω is a countable set and $\mathcal{F}$ consists of all subsets of Ω. Show that $X_n \to X$ in probability implies $X_n \to X$ a.s.

Proof. Suppose for contradiction that X_n does not converge to X a.s. Then there is a set $A \subset \Omega$ such that $P(A) > 0$ and $X_n(\omega)$ does not converge to $X(\omega)$ for all $\omega \in A$. Let

$$B_r = \left\{\omega : |X_n(\omega) - X(\omega)| > 2^{-r} \text{ i.o.}\right\},$$

then $A = \bigcup_{r=1}^{\infty} B_r$. By continuity of measure, there exists r such that $P(B_r) = \alpha > 0$ Since B_r is countable, enumerate the elements in B_r by $\{\omega_i\}_{i=1}^{\infty}$. Write $\varepsilon = 2^{-r}$.

Now, there is a subsequence $n^1(j)$ such that $\left|X_{n^1(j)}(\omega_1) - X(\omega_1)\right| > \varepsilon$ for all j. Once we construct $n^{k-1}(j)$, there is a subsequence $n^k(j)$ such that $\left|X_{n^k(j)}(\omega_k) - X(\omega_k)\right| > \varepsilon$ for all j. Let $m(k) = n^k(k)$, then $\left|X_{m(k)}(\omega) - X(\omega)\right| > \varepsilon$ for all k and all $\omega \in A$. This implies

$$P\left(\left|X_{m(k)} - X(\omega)\right| > \varepsilon\right) = \alpha > 0.$$

Hence, $P(|X_n - X| > \varepsilon) = \alpha$ i.o. and X_n does not converge to X in probability, contradiction.

□

Exercise (Durrett 2.3.14)**.** Let $X_1, X_2, \dots$ be independent. Show that $\sup X_n < \infty$ a.s. if and only if $\sum_{n=1}^{\infty} P(X_n > A) < \infty$ for some A.

Proof. ($\Longrightarrow$) Suppose for contradiction that $\sum_{n=1}^{\infty} P(X_n > A) = \infty$ for any $A < \infty$. By independence and the second Borel-Cantelli lemma, we know $P(X_n > A \text{ i.o.}) = 1$ for any A. Since A is arbitrary, this then implies $\sup X_n = \infty$ a.s., contradiction.

($\Longleftarrow$) Suppose $\sum_{n=1}^{\infty} P(X_n > A) < \infty$ for some A. Then, Borel-Cantelli lemma implies $P(X_n > A \text{ i.o.}) = 0$. That is, for almost sure $\omega \in \Omega$, there exists $N(\omega)$ such that $X_n(\omega) \leq A$ for all $n \geq N$. For these ω, $\sup X_n(\omega) < \infty$. Therefore, $\sup X_n < \infty$ a.s.

□

3 Central Limit Theorems

3.3 Characteristic functions

Exercise (Durrett 3.3.10)**.** Using the identity $\sin t = 2\sin\frac{t}{2}\cos\frac{t}{2}$ repeatedly leads to

$$\frac{\sin t}{t} = \prod_{m=1}^{\infty}\cos\left(\frac{t}{2^m}\right).$$

Prove the last identity by interpreting each side as a characteristic function.

Proof. Note that

$$\frac{1}{2}\int_{-1}^{1} e^{itx}\,dx = \frac{\sin t}{t}.$$

Therefore, LHS can be interpreted as the ch.f. of a unif$[-1,1]$ random variable. Meanwhile,

$$\cos\left(2^{-m}t\right) = \frac{1}{2}\exp(it2^{-m}) + \frac{1}{2}\exp(-it2^{-m}).$$

Hence, $\cos(2^{-m}t)$ can be interpreted as ch.f. of a random variable X_m that maps to $\pm 2^{-m}$ with probability 1/2 each, and RHS can be interpreted as the sum of independent X_m's. Now we just need to show that $X_1 + \cdots + X_n \Rightarrow$ unif$[-1,1]$. To verify this, note that $X_m = 2^{-m}(2b_m - 1)$ where $\{b_m\}_{m=1}^{\infty}$ are i.i.d. Bernoulli(1/2) random variables. Meanwhile, $\sum_{m=1}^{\infty} 2^{-m}b_m$ converges to unif$[0,1]$ by interpreting the sum as choosing binary digits. It then immediately follows that $X_1 + \cdots + X_n \Rightarrow$ unif$[-1,1]$. More concretely, we can compute the probability of each dyadic intervals and approximate all intervals through them. This then uniquely determines the distribution. By Levy's theorem, the proof for the desired inequality is complete.

□

Exercise (Durrett 3.3.20)**.** If $X_1, X_2, \ldots$ are independent and have characteristic function $\exp(-|t|^{\alpha})$ then $(X_1 + \cdots + X_n)/n^{1/\alpha}$ has the same distribution as X_1.

Proof. By independence, the ch.f. of $X_1 + \cdots + X_n$ is $\exp(-n|t|^{\alpha})$. Hence the ch.f. of $(X_1 + \cdots + X_n)/n^{1/\alpha}$ is

$$\exp\left(-n\left|\frac{t}{n^{1/\alpha}}\right|^{\alpha}\right) = \exp(-|t|^{\alpha}).$$

This is the same ch.f. as X_1, so $(X_1 + \cdots + X_n)/n^{1/\alpha}$ has the same distribution as X_1.

□

Exercise (Durrett 3.3.23)**.** Show that if X and Y are independent and $X + Y$ and X have the same distribution then $Y = 0$ a.s.

Proof. Since X and Y are independent, $\varphi_{X+Y} = \varphi_X \varphi_Y$. We are given that $\varphi_X \varphi_Y = \varphi_X$, so

$$\varphi_X(t)(\varphi_Y(t) - 1) = 0$$

for all $t \in \mathbb{R}$. Note that $\varphi_X(0) = 1$ and φ_X is continuous, so $\varphi_Y = 0$ on a neighborhood of zero. Now show that this implies $Y = 0$ a.s.

Let t be in this neighborhood, then $1 = E[\cos(tY)]$. Note that $\cos(tY) \leq 1$, so this implies $\cos(tY) = 1$ a.s., and $tY \in 2\pi\mathbb{Z}$ a.s. Let t_1, t_2 be both in this neighborhood and $t_1/t_2 \notin \mathbb{Q}$. Then $t_1 Y \in 2\pi\mathbb{Z}$ and $t_2 Y \in 2\pi\mathbb{Z}$ a.s. If $Y(\omega) \neq 0$, then

$$\frac{t_1}{t_2} = \frac{t_1 Y(\omega)}{t_2 Y(\omega)} \in \mathbb{Q},$$

a contradiction. Therefore, $Y = 0$ a.s.

□

4 Martingales

4.1 Conditional expectation

Exercise (Durrett 4.1.2)**.** Prove **Chebyshev's inequality**. If $a > 0$ then

$$P\{|X| \geq a|\mathcal{F}\} \leq a^{-2}E[X^2|\mathcal{F}].$$

Proof. Let $A \in \mathcal{F}$, we have

$$\int_A E[\mathbf{1}_{\{|X|\geq a\}}\,|\mathcal{F}]\,dP = \int_A \mathbf{1}_{\{|X|\geq a\}}\;dP.$$

On the event $\{|X| \geq a\}$, $a^{-2}X^2 \geq 1$, so

$$\begin{aligned}\int_A E[\mathbf{1}_{\{|X|\geq a\}}\,|\mathcal{F}]\,dP &\leq \int_A a^{-2}X^2\,dP \\ &= a^{-2}\int_A X^2\,dP \\ &= a^{-2}\int_A E[X^2|\mathcal{F}]\,dP.\end{aligned}$$

Since $A \in \mathcal{F}$ is arbitrary, this completes the proof.

□

Exercise (Durrett 4.1.5)**.** Give an example on $\Omega = \{a, b, c\}$ in which

$$E[E[X|\mathcal{F}_1]|\mathcal{F}_2] \neq E[E[X|\mathcal{F}_2]|\mathcal{F}_1].$$

Proof. Let P be the uniform distribution and $X = \mathbf{1}_{\{a\}}$. Also let

$$\mathcal{F}_1 = \{\{a,b\},\{c\},\emptyset,\Omega\}, \quad \mathcal{F}_2 = \{\{b,c\},\{a\},\emptyset,\Omega\}.$$

Then $E[X|\mathcal{F}_2] = X$. Meanwhile, $E[X|\mathcal{F}_1](c) = 0$ and $E[X|\mathcal{F}_1](a) = E[X|\mathcal{F}_1](b) = \frac{1}{2}$. With this, we can also compute $E[E[X|\mathcal{F}_1]|\mathcal{F}_2](a) = \frac{1}{2}$ and $E[E[X|\mathcal{F}_1]|\mathcal{F}_2](b) = E[E[X|\mathcal{F}_1]|\mathcal{F}_2](c) = \frac{1}{4}$. This shows that $E[E[X|\mathcal{F}_1]|\mathcal{F}_2] \neq E[E[X|\mathcal{F}_2]|\mathcal{F}_1]$, as desired.

□

Exercise (Durrett 4.1.6)**.** Show that if $\mathcal{G} \subset \mathcal{F}$ and $EX^2 < \infty$ then

$$E\left[(X - E[X|\mathcal{F}])^2\right] + E\left[(E[X|\mathcal{F}] - E[X|\mathcal{G}])^2\right] = E\left[(X - E[X|\mathcal{G}])^2\right].$$

Proof. For LHS, we have

$$
\begin{aligned}
&(X - E[X|\mathcal{F}])^2 + (E[X|\mathcal{F}] - E[X|\mathcal{G}])^2 \\
&= X^2 + 2E[X|\mathcal{F}]^2 - 2XE[X|\mathcal{F}] - 2E[X|\mathcal{F}]E[X|\mathcal{G}] + E[X|\mathcal{G}]^2
\end{aligned}
$$

For RHS, we have

$$(X - E[X|\mathcal{G}])^2 = X^2 + E[X|\mathcal{G}]^2 - 2XE[X|\mathcal{G}].$$

Hence, we just need to show

$$E[E[X|\mathcal{F}]^2 - XE[X|\mathcal{F}] - E[X|\mathcal{F}]E[X|\mathcal{G}]] = E[-XE[X|\mathcal{G}]].$$

Using the tower property and taking conditional expectation w.r.t. $\mathcal{F}$ first and then taking the unconditional expectation, we have

$$
\begin{aligned}
&E[E[X|\mathcal{F}]^2 - XE[X|\mathcal{F}] - E[X|\mathcal{F}]E[X|\mathcal{G}]] \\
&= E[E[X|\mathcal{F}]^2] - E[E[X|\mathcal{F}]^2] - E[E[X|\mathcal{F}]E[X|\mathcal{G}]] \\
&= -E[E[X|\mathcal{F}]E[X|\mathcal{G}]],
\end{aligned}
$$

Similarly, we have

$$E[-XE[X|\mathcal{G}]] = -E[E[X|\mathcal{F}]E[X|\mathcal{G}]],$$

where we used the fact that $E[X|\mathcal{G}]$ is $\mathcal{G}$-measurable and thus $\mathcal{F}$-measurable.

□

Exercise (Durrett 4.1.8)**.** Let $Y_1, Y_2, \ldots$ be i.i.d. with mean μ and variance σ^2, N an independent positive integer valued r.v. with $EN^2 < \infty$ and $X = Y_1 + \cdots + Y_N$. Show that

$$\operatorname{var}(X) = \sigma^2 EN + \mu^2 \operatorname{var}(N).$$

To understand and help remember the formula, think about the two special cases in which N or Y is constant.

Proof. It is easy to see that $E[Y_1^2] = \sigma^2 + \mu^2$. Then, we have

$$E[X] = E[E[X|N]] = E[\mu N] = \mu EN,$$

and

$$E[X^2] = E[E[X^2|N]] = E[N(\sigma^2 + \mu^2) + N(N-1)\mu^2] = \mu^2 E[N^2] + \sigma^2 EN.$$

Subtrating these two expressions gives $\operatorname{var} X = \sigma^2 EN + \mu^2 \operatorname{var} N$, as desired. □

Exercise (Durrett 4.1.9)**.** Show that if X and Y are random variables with $E[Y|\mathcal{G}] = X$ and $EY^2 = EX^2 < \infty$, then $X = Y$ a.s.

Proof. Since $X = E[Y|\mathcal{G}]$, X is $\mathcal{G}$-measurable. It follows that

$$E[(Y-X)X|\mathcal{G}] = E[(Y-X)X|\mathcal{G}] = X(E[Y|\mathcal{G}] - E[X]) = X(X-X) = 0,$$

so $E[(Y-X)X] = 0$ i.e. $E[XY] = E[X^2]$. Now we have

$$E[(Y-X)^2] = E[Y^2] - 2E[XY] + E[X^2] = E[Y^2] - E[X^2] = 0.$$

Therefore, $X = Y$ a.s. □

4.2 Martingales, almost sure convergence

Exercise (Durret 4.2.1)**.** Suppose X_n is a martingale w.r.t. $\mathcal{G}_n$ and let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Then $\mathcal{G}_n \supset \mathcal{F}_n$ and X_n is a martingale w.r.t $\mathcal{F}_n$.

Proof. By assumption, $X_1, \dots, X_n$ are all $\mathcal{G}_n$ measurable. Therefore it is clear that $\mathcal{G}_n \supset \mathcal{F}_n$. Moreover,

$$E[X_{n+1}|\mathcal{F}_n] = E[E[X_{n+1}|\mathcal{G}_n]|\mathcal{F}_n] = E[X_n|\mathcal{F}_n] = X_n,$$

so X_n is a martingale w.r.t. $\mathcal{F}_n$.

□

Exercise (Durret 4.2.5)**.** Give an example of a martingale X_n with $X_n \to -\infty$ a.s. Hint: Let $X_n = \xi_1 + \cdots + \xi_n$ where the ξ_i are independent (but not identically distributed) with $E\xi_i = 0$.

Proof. Define ξ_n via $P\{\xi_n = -n^{-1}\} = 1 - n^{-2}$ and $P\{\xi_n = n - n^{-1}\} = n^{-2}$. Then

$$E\xi_n = -\left(1 - \frac{1}{n^2}\right)\frac{1}{n} + \frac{1}{n^2}\left(n - \frac{1}{n}\right) = 0.$$

Therefore $X_n = \xi_1 + \cdots + \xi_n$ is a martingale. To show $X_n \to -\infty$ a.s., note that

$$\sum_{i=1}^{\infty} P\{\xi_n = n - n^{-1}\} = \sum_{i=1}^{\infty} \frac{1}{n^2} < \infty.$$

Borel-Cantelli lemma then implies that $P\{\xi_n = n - n^{-1}$ i.o.$\} = 0$. On the complementary event $\{\xi_n = -n^{-1}$ for all but finitely many $n\}$, it is clear that $X_n \to -\infty$.

□

Exercise (Durret 4.2.8)**.** Let X_n and Y_n be positive integrable and adapted to $\mathcal{F}_n$. Suppose

$$E[X_{n+1}|\mathcal{F}_n] \le (1 + Y_n)X_n$$

with $\sum Y_n < \infty$ a.s. Prove that X_n converges a.s. to a finite limit by finding a closely related supermartingale to which Theorem 4.2.12 can be applied.

Proof. Define $A_n = \prod_{k=1}^{n-1}(1+Y_k)$ and $Z_n = \frac{X_n}{A_n}$. Since Y_k is $\mathcal{F}_k$-measurable for each k, A_n is $\mathcal{F}_n$-measurable. Also, since $A_n \geq 1$ a.s. and X_n is integrable, Z_n is also integrable. Now we have $A_{n+1} = A_n(1+Y_n)$, and thus

$$E[Z_{n+1}|\mathcal{F}_n] = E\left[\frac{X_{n+1}}{A_n(1+Y_n)}\middle|\mathcal{F}_n\right] = \frac{E[X_{n+1}|\mathcal{F}_n]}{A_n(1+Y_n)} \leq \frac{X_n}{A_n} = Z_n.$$

Therefore, Z_n is a supermartingale and thus converges to some finite limit Z a.s.

Now consider A_n. Using $\log(1+x) \leq x$ for $x > 0$, we have

$$\sum_{k=1}^{n-1} \log(1+Y_k) \leq \sum_{k=1}^{n-1} Y_n \leq \sum_{n=1}^{\infty} Y_n < \infty \text{ a.s.}$$

Therefore, the series $\sum \log(1+Y_k)$ converges a.s. and thus $A_n = \exp(\sum_{k=1}^{n-1} \log(1+Y_k))$ converges to a finite limit A a.s. This implies $X_n \to ZA$ a.s.

□

Exercise (Durret 4.2.9)**. The switching principle**. Suppose X_n^1 and X_n^2 are supermartingales with respect to $\mathcal{F}_n$ and N is stopping time so that $X_N^1 \geq X_N^2$. Then

$$\begin{aligned} Y_n &= X_n^1 \,\mathbf{1}_{\{N>n\}} + X_n^2 \,\mathbf{1}_{\{N\leq n\}} \text{ is a supermartingale.} \\ Z_n &= X_n^1 \,\mathbf{1}_{\{N\geq n\}} + X_n^2 \,\mathbf{1}_{\{N<n\}} \text{ is a supermartingale.} \end{aligned}$$

Proof. To show Y_n is a supermartingale, note that

$$Y_{n+1} = X_{n+1}^1 \,\mathbf{1}_{\{N>n+1\}} + X_{n+1}^2 \,\mathbf{1}_{\{N\leq n+1\}}$$

However, $\mathbf{1}_{\{N>n+1\}} + \mathbf{1}_{\{N=n+1\}} = \mathbf{1}_{\{N>n\}}$ and $\mathbf{1}_{\{N\leq n+1\}} = \mathbf{1}_{\{N\leq n\}} + \mathbf{1}_{\{N=n+1\}}$. It follows that

$$Y_{n+1} = X_{n+1}^1 \,\mathbf{1}_{\{N>n\}} + X_{n+1}^2 \,\mathbf{1}_{\{N\leq n\}} + (X_{n+1}^2 - X_{n+1}^1) \,\mathbf{1}_{\{N=n+1\}}.$$

Note that $\mathbf{1}_{\{N\leq n\}}$ and $\mathbf{1}_{\{N>n\}}$ are both $\mathcal{F}_n$-measurable, and $X_{n+1}^1 \geq X_{n+1}^2$ on the event $\{N = n+1\}$. Taking the conditional expectation w.r.t. $\mathcal{F}_n$ gives

$$\begin{aligned} E[Y_{n+1}] &\leq \mathbf{1}_{\{N>n\}} \, E[X_{n+1}^1|\mathcal{F}_n] + \mathbf{1}_{\{N\leq n\}} \, E[X_{n+1}^2|\mathcal{F}_n] \\ &\leq \mathbf{1}_{\{N>n\}} \, X_n^1 + \mathbf{1}_{\{N\leq n\}} \, X_{n+1}^2 \\ &= Y_n, \end{aligned}$$

showing Y_n is a supermartingale.

To show Z_n is a supermartingale, note that $\mathbf{1}_{\{N \geq n+1\}}$ and $\mathbf{1}_{\{N<n+1\}}$ are $\mathcal{F}_n$-measurable. It follows that

$$\begin{aligned} E[Z_{n+1}|\mathcal{F}_n] &= E[X^1_{n+1}\,\mathbf{1}_{\{N \geq n+1\}} + X^2_{n+1}\,\mathbf{1}_{\{N<n+1\}}\,|\mathcal{F}_n] \\ &= E[X^1_{n+1}|\mathcal{F}_n]\,\mathbf{1}_{\{N \geq n+1\}} + E[X^2_{n+1}|\mathcal{F}_n]\,\mathbf{1}_{\{N<n+1\}} \\ &\geq X^1_n\,\mathbf{1}_{\{N \geq n+1\}} + X^2_n\,\mathbf{1}_{\{N<n+1\}}\,. \end{aligned}$$

Now we can use a similar decomposition to see $E[Z_{n+1}|\mathcal{F}_n] \geq Z_n$.

□